

EXHIBIT 34

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U.S. Patent No. 9,093,425 Infringement Chart

Title: Self-aligned liner formed on metal semiconductor alloy contacts

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Abstract: Metal semiconductor alloy contacts are provided on each of a source region and a drain region which are present in a semiconductor substrate. A transition metal is then deposited on each of the metal semiconductor alloy contacts, and during the deposition of the transition metal, the deposited transition metal reacts preferably, but not necessarily always, in-situ with a portion of each the metal semiconductor alloy contacts forming a transition metal-metal semiconductor alloy liner atop each metal semiconductor alloy contact. Each transition metal-metal semiconductor alloy liner that is provided has outer edges that are vertically coincident with outer edges of each metal semiconductor alloy contact. The transition metal-metal semiconductor alloy liner is more etch resistant as compared to the underlying metal semiconductor alloy. As such, the transition metal-metal semiconductor alloy liner can serve as an effective etch stop layer during any subsequently performed etch process.

Claim	Tower '425 Accused Products
[1pre] A semiconductor structure comprising:	To the extent the preamble is limiting, devices manufactured by Tower that rely on its RF processes (including its 65 nm RF process) (hereinafter, "Tower '425 Accused Products") comprise a semiconductor structure. For example, the Tower '425 Accused Products include the Qorvo QM81026 chip.¹ That is, on information and belief, the QM81026 has substantially similar structure, function, operation, and implementation with respect to the claims of the '425 patent as any other chip manufactured by Tower using its 65nm process.
[1a] a semiconductor substrate having a source region and a drain region located within a semiconductor material portion of said	The Tower '425 Accused Products comprise a semiconductor substrate having a source region and a drain region located within a semiconductor material portion of said semiconductor substrate, wherein said source region and said drain region are spaced apart by a channel region. For example, the Tower '425 Accused Products include a semiconductor substrate with a channel region between a source region and a drain region, as shown in the TEM image below.

¹ GlobalFoundries reserves the right to identify to identify different or additional Tower '425 Accused Products and amend or supplement these charts after further investigation and discovery, claim construction, or other meriting circumstances. Such supplemental information may include, but is not limited to, source code, data sheets, design specifications, deposition testimony, testing information, reference designs implementation and utilization information, and/or schematics.

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Claim	Tower '425 Accused Products
semiconductor substrate, wherein said source region and said drain region are spaced apart by a channel region;	
[1b] a functional gate structure located above said channel region;	The Tower '425 Accused Products comprise a functional gate structure located above said channel region. For example, the '425 Accused Products such as Qorvo QM81026 chip include a functional gate structure located above said channel region.

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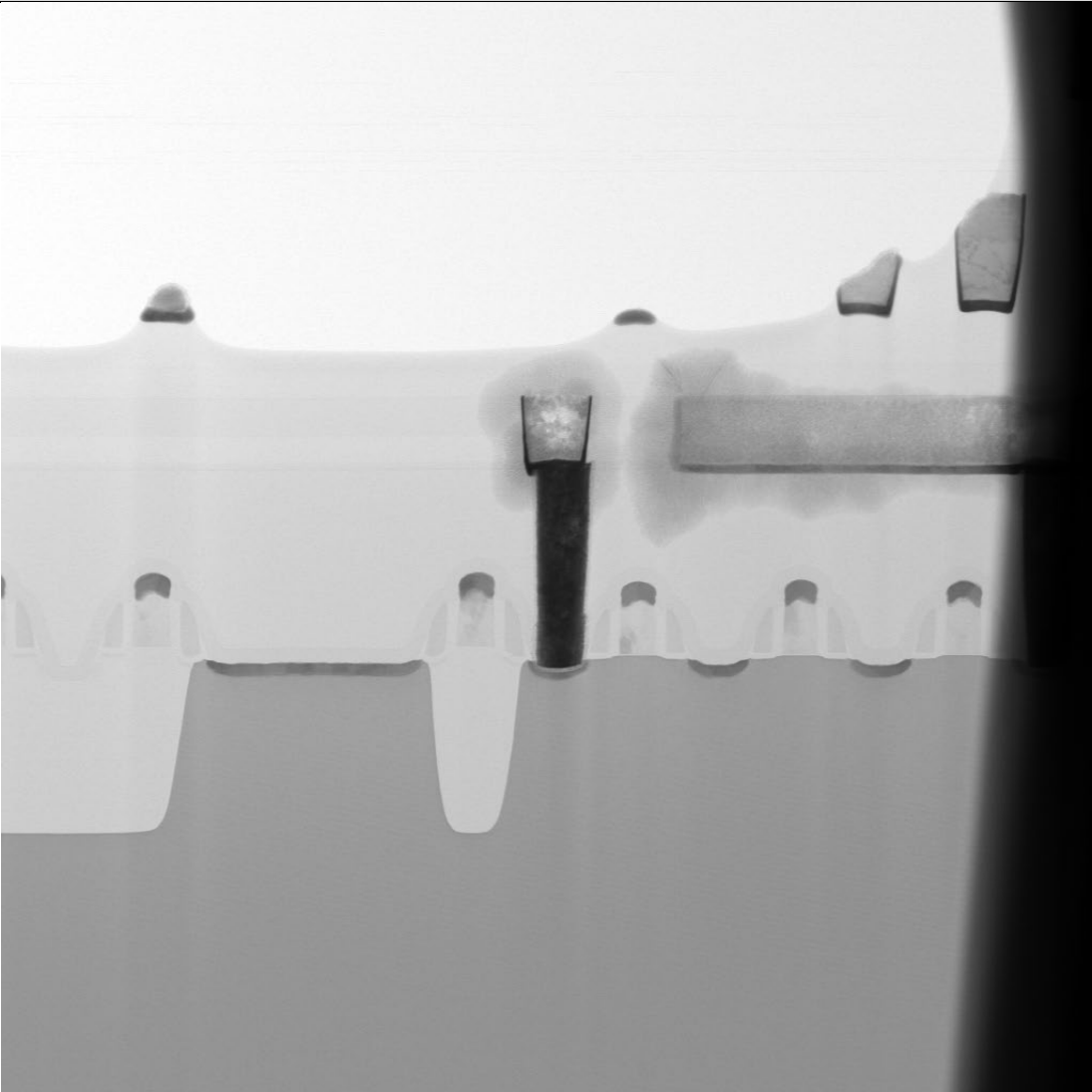
Claim	Tower '425 Accused Products
	 STEM Mag 51.1 kx MAL-00-26 iPhone16e QM81026 CA on RX 200 nm BF-S

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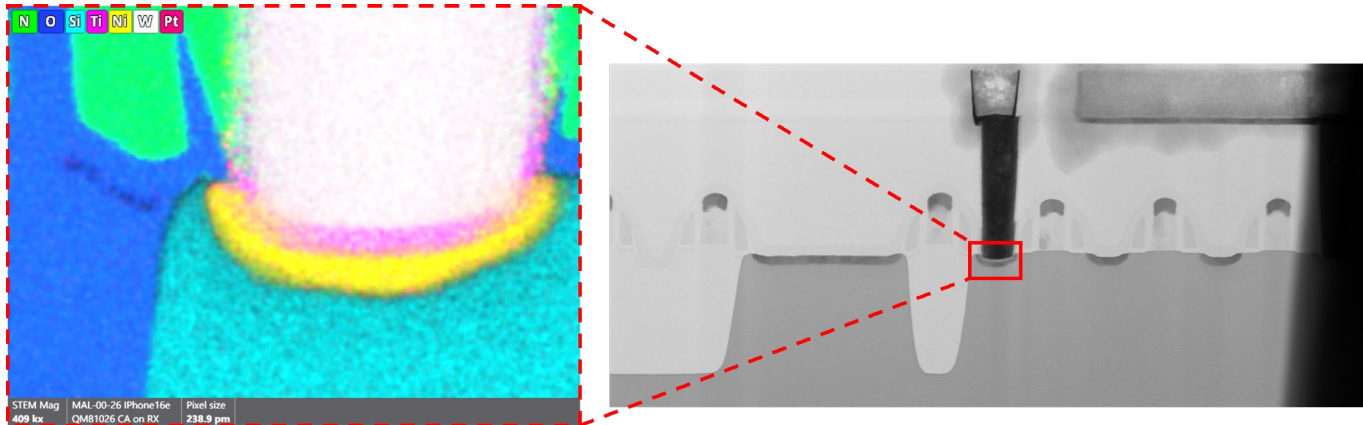
Claim	Tower '425 Accused Products
[1c] a source-side material stack of, from bottom to top, a source-side metal semiconductor alloy portion and a source-side transition-metal metal semiconductor alloy portion located on one side of said functional gate structure and located atop said source region; and	The Tower '425 Accused Products comprise a source-side material stack of, from bottom to top, a source-side metal semiconductor alloy portion and a source-side transition-metal metal semiconductor alloy portion located on one side of said functional gate structure and located atop said source region. For example, the source-side material stack in the Tower '425 Accused Products includes a metal semiconductor alloy portion below a transition-metal metal semiconductor alloy portion. The images below show the source-side material stack.  STEM Map MAL-00-26 iPhone16e Pixel size 409 kx QM81026 CA on RX 238.9 µm The green arrow in the image below corresponds to the green arrow in the following graph (i.e., such that the downward direction in the image below corresponds to the left-to-right direction in the ensuing graph).

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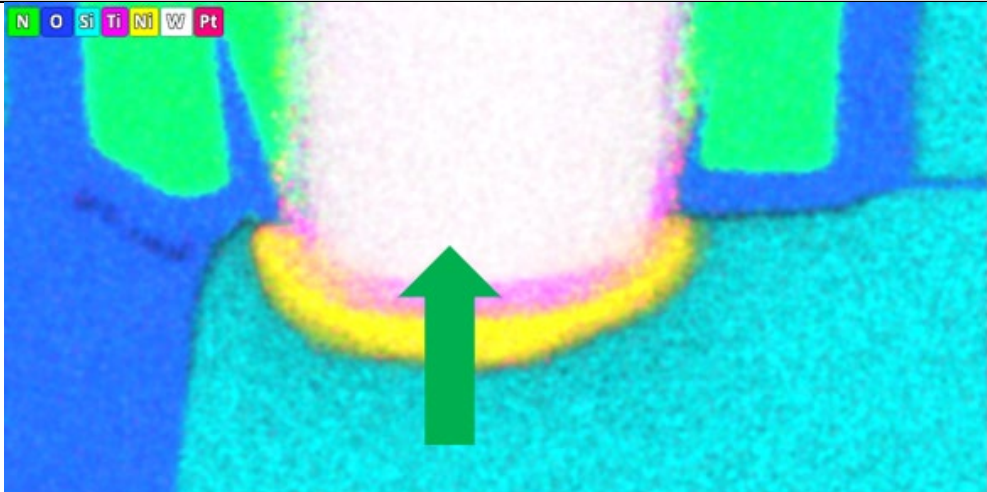
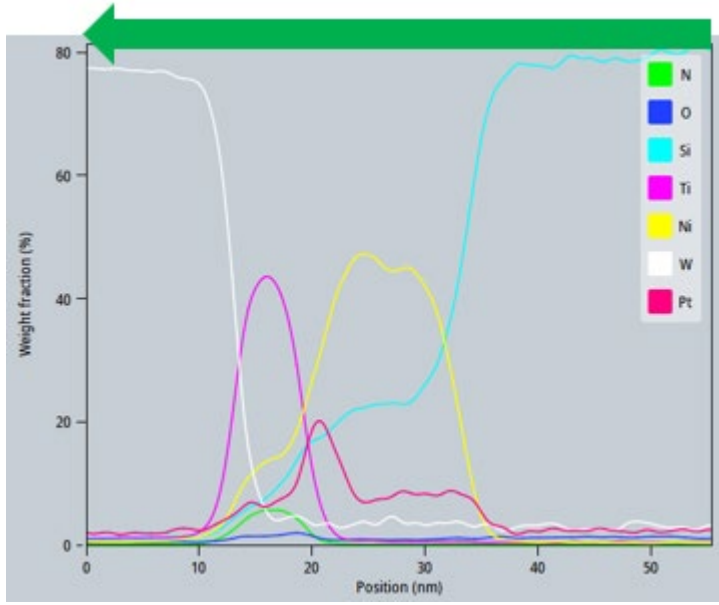
Claim	Tower '425 Accused Products
	An EDS spectrum image showing a cross-section of a material stack. The image is color-coded by elemental composition. A legend at the top left shows color swatches for N (green), O (blue), Si (cyan), Ti (magenta), Ni (yellow), W (white), and Pt (red). The image shows a central white region, likely Pt, surrounded by a yellow/orange layer, which is further surrounded by a blue layer. A green arrow points upwards from the bottom towards the yellow/orange layer. As shown, the material stack comprises nickel silicide and a titanium, nickel, silicon alloy.

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Claim	Tower '425 Accused Products
	
[1d] a drain-side material stack of, from bottom to top, a drain-side metal semiconductor alloy portion and a drain-side transition metal-metal semiconductor alloy portion located on another side of said functional gate structure and located atop said drain region.	The Tower '425 Accused Products comprise a drain-side material stack of, from bottom to top, a drain-side metal semiconductor alloy portion and a drain-side transition metal-metal semiconductor alloy portion located on another side of said functional gate structure and located atop said drain region. The drain-side material stack has the same composition as the source-side material stack. <i>See [1c].</i>